

	$g = \frac{u^2}{R_0}$ $= \frac{(12.6)^2}{16.3}$ $= 9.7399 \text{ m s}^{-2}$	C1
	$\frac{\Delta g}{g} = 2 \frac{\Delta u}{u} + \frac{\Delta R_0}{R_0}$	A1

	$\frac{\Delta g}{9.7399} = 2\left(\frac{3}{100}\right) + \frac{4}{100}$ $\Delta g = 0.97$ $= 1 \text{ m s}^{-2} \text{ (to 1 s.f.)}$ Therefore, $g = (10 \pm 1) \text{ m s}^{-2}$	A1 A1
	<i>Marker's comments: Uncertainty should only be left to 1 sig fig. The final answer and its uncertainty should have the same d.p. It is important to convert the speed to m s^{-1} to obtain g in units of m s^{-2}. Also, there were mistakes in understanding the meaning of percentage uncertainty.</i>	
2(a)	The upthrust is the <u>resultant upward</u> force due to the <u>larger</u> pressure exerted by the fluid at the bottom compared to the top surface of the body.	B1
2(b)	$\text{Upthrust} = W_{\text{displaced fluid}}$ $= \rho_{\text{fluid}} V_{\text{cannon}} g$ $= 1050 \left(\frac{800}{8000} \right) (9.81)$ $= 1030 \text{ N (shown)}$	M1 A0
	<i>Marker's comments: It is a "show" type of question and thus important that all steps, including substitution must be explained and shown.</i>	
2(c)	The rate of change of momentum of a body is directly proportional to the <u>resultant</u> force acting on the body and occurs in the direction of the <u>resultant</u> force.	B1
2(d)	During its upward motion, the cannon experiences viscous <u>drag force</u> which acts in the opposite direction of its motion and <u>increases with the speed of the cannon</u>. Net force on canon = upthrust – (weight + drag force) <u>decreases</u> (since weight and upthrust are constant) Since the net upward force decreases over time, the momentum increases non-linearly (since the net force is directly proportional to the rate of change of momentum)	B1 B1
	<i>Marker's comments: Very often, drag force is not mentioned. There were misconception that the upthrust decreases causing net force to decrease or the velocity decreases while the momentum increases.</i>	

2(e)	$\begin{aligned}\text{gradient of tangent line} &= F_{\text{net}} \\ &= \frac{400 - 0}{4 - 0} \\ &= 100 \text{ N}\end{aligned}$ (M1 for tangent line and computing gradient correctly) Net force on cannon = upthrust of cannon and bag – weight of cannon $\begin{aligned}F_{\text{net}} &= \underline{U} - W \\ 100 &= U - (800)(9.81) \\ U &= 7948 \text{ N}\end{aligned}$ Hence upthrust on bag = 7948 – 1030 = 6918 N For the bag, $\underline{U} = V\rho g$ $6918 = V(1050)(9.81)$ $V = 0.672 \text{ m}$	M1 C1 A1
2(e)	Marker's comments: Poorly attempted. Gradient not drawn to obtain the initial net force.	D1